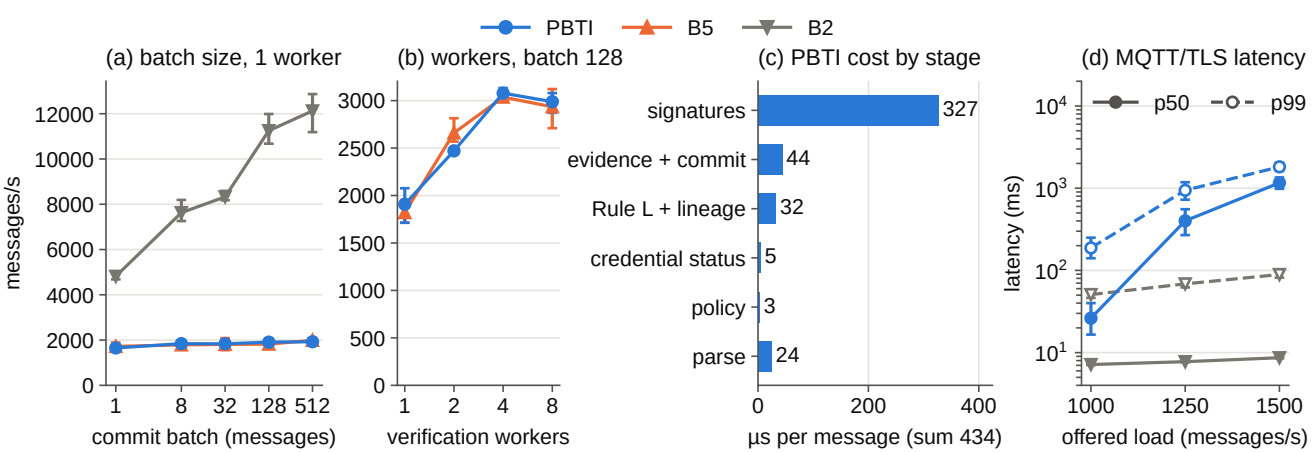
PBTI
B5
B2
(a) batch size, 1 worker
messages/s
12000
10000
8000
6000
4000
2000
0
1
8
32
128
512
commit batch (messages)
(b) workers, batch 128
3000
2500
2000
1500
1000
500
0
1
2
4
8
verification workers
(c) PBTI cost by stage
signatures
evidence + commit
Rule L + lineage
credential status
policy
parse
327
44
32
5
3
24
0
200
400
µs per message (sum 434)
(d) MQTT/TLS latency
p50
p99
latency (ms)
10^4
10^3
10^2
10^1
1000
1250
1500
offered load (messages/s)